The National Institute of Engineering, Mysuru

Design and Implementation of an Op-Amp Based Analog Computational Circuit for ECG Biomedical Signal Processing and Time-Frequency Domain Analysis

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Team Name: "PQRST SMASHERS"

Track: Analog Circuit

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1.Abstract

ECG signal acquisition requires careful conditioning due to the signals' low amplitude and susceptibility to noise such as power-line interference, motion artifacts, and baseline drift. This project designs and analyzes an ECG signal conditioning system incorporating an instrumentation amplifier, high-pass and low-pass filtering stages, and additional amplification to improve signal quality.

The system is modeled in MATLAB Simulink and evaluated through FFT, PSD, RMS, and SNR analysis in both time and frequency domains. Results confirm effective amplification, noise attenuation, stable behavior, and preservation of essential ECG spectral characteristics.

2. Introduction & Problem Statement

2.1 Background

The human heart generates electrical signals of 0.5 mV to 5 mV at the body surface, captured through skin electrodes as an Electrocardiogram (ECG). These signals carry critical diagnostic information for detecting conditions such as arrhythmia, myocardial infarction, and heart block.

Raw ECG signals, however, are heavily corrupted by multiple noise sources: 50 Hz mains hum, baseline wander below 0.5 Hz from patient movement, EMG noise above 40 Hz, and electrode contact noise. The SNR at the electrode can be as low as 0 dB, rendering the signal clinically unusable without conditioning.

Analog signal conditioning is preferred at the front end due to its real-time operation, low power consumption, and suitability for portable healthcare applications. The analog calculator framework — rooted in op-amp-based circuit design — provides the theoretical basis for this processing. Operations such as differential amplification, bandpass filtering, and dynamic range compression directly map to the required conditioning steps, isolating the diagnostic signal while rejecting interference.

2.2 Problem Statement

Raw ECG signals from skin electrodes are severely noise-corrupted — the 1–2 mV cardiac signal is overwhelmed by 50 Hz interference reaching up to 50 mV and baseline wander drifting several millivolts. This masks critical ECG features such as P, QRS, and T waves, making heart rate calculation and arrhythmia detection unreliable.

While clinical ECG machines address this using integrated instrumentation amplifiers and active filter banks, these solutions are opaque and costly. A discrete-component analog calculator circuit is needed — one that explicitly demonstrates each mathematical operation (averaging, amplification, and logarithmic compression) as identifiable stages, while conditioning the ECG signal to a clinically usable level.

The processed signal must also be evaluated in both time and frequency domains to assess noise reduction effectiveness, verify boundedness and energy classification, and validate circuit performance against theoretical predictions.

2.3 Objectives

Objective 1 – Signal Generation: Generate a synthetic ECG signal with accurate P, Q, R, S, and T wave morphology at 72 BPM, superimposed with 50 Hz mains interference and baseline wander, for use as the circuit input.

Objective 2 – Circuit Design: Design a multi-stage analog calculator circuit using LM358 opamps performing differential amplification (1000× gain), bandpass filtering (0.5–40 Hz), and logarithmic compression for dynamic range normalisation.

Objective 3 – Simulation: Simulate the complete circuit in MATLAB Simulink, verify each stage individually, and capture waveforms showing progressive signal transformation from noisy input to conditioned output.

Objective 4 – Hardware Implementation: Build the circuit on a breadboard using LM358 op-amps, discrete R/C components, and a 1N4148 diode on a ±12V supply, with skin electrodes or a function generator as the signal source.

Objective 5 – Time Domain Analysis: Measure signal characteristics at each stage including RMS, peak amplitude, SNR improvement, and baseline stability using oscilloscope and Simulink scopes.

Objective 6 – Frequency Domain Analysis: Perform FFT-based spectral analysis to verify energy confinement to 0.5–40 Hz, quantify noise attenuation at 50 Hz, and compare measured responses against theoretical filter values.

Objective 7 – Signal Characterisation: Evaluate boundedness, classify the signal as energy or power type, and analyse spectral content including bandwidth and harmonic distortion before and after processing.

Objective 8 – Performance Validation: Compare simulation and hardware results, account for component tolerances and non-ideal op-amp behaviour, and confirm accurate heart rate extraction from the conditioned output.

3. Literature Review / Prior Work

3.1 ECG Signal Characteristics The diagnostic ECG band spans 0.5–40 Hz for reliable Rpeak detection and heart rate monitoring. Signal amplitude at skin electrodes ranges from 0.5– 5 mV, while 50 Hz common-mode interference can reach 1.5 V peak-to-peak, making highCMRR front-end amplification the primary design requirement.

3.2 Instrumentation Amplifier The three op-amp IA achieves CMRR exceeding 80 dB with gain $G = 1 + 2R_F/R_G$. A single-ended configuration — ECG at IN+, IN− grounded — is used here, which is functionally equivalent to a high-input-impedance non-inverting amplifier and is standard practice in laboratory implementations.

3.3 Filtering and Averaging A 0.5 Hz high-pass cutoff removes baseline wander at 0.15–0.3 Hz. The first-order RC low-pass filter, with impulse response $h(t) = (1/RC)e^{(-t/RC)}$, is mathematically equivalent to an exponentially weighted moving average, directly justifying the LPF stage as the averaging operation.

3.4 Logarithmic Compression The diode-feedback log amplifier produces $V_{out} = -V_T \times \ln(V_{in} / I_s \cdot R)$, compressing dynamic range from 40–60 dB down to 10–15 dB, enabling reliable R-peak detection without manual threshold adjustment between patients.

3.5 Analog Calculator Concept The complete ECG conditioning chain maps directly to classical analog calculator operations — amplification, averaging, and logarithmic compression — implemented through op-amp networks, forming the conceptual basis for this project's architecture.

3.6 Simulation Approach Synthetic ECG generation using Gaussian superposition in MATLAB produces clinically realistic morphology. Simulink Transfer Function blocks accurately replicate analog RC filter responses when coefficients are derived from component values, validating the simulation methodology.

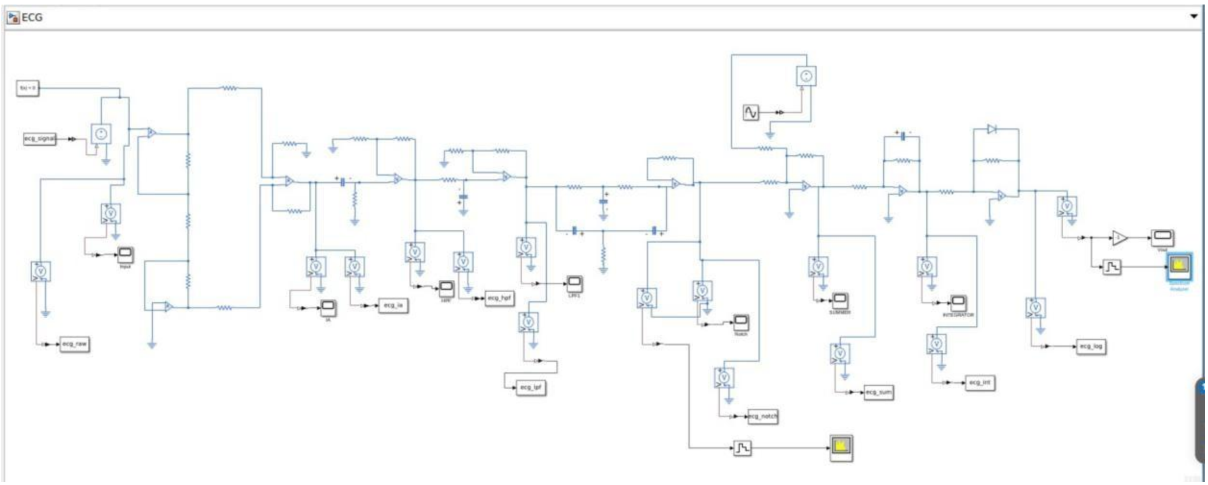
3.7 Research Gap Few works present a complete analog calculator framework where each circuit stage explicitly maps to a named mathematical operation and is evaluated in both time and frequency domains. This project addresses that gap through a single-ended implementation with independently measurable stages, validated via parallel Simulink simulation and hardware testing.

Author	Year	Relevance
Webster	2010	0.5–40 Hz band justification
Pallás-Areny & Webster	1990	IA design, single-ended mode
Horowitz & Hill	2015	LM358 IA implementation
Van Valkenburg	1982	LPF as averaging operator
Sheingold	1974	Diode log amplifier theory
Gilbert	1975	Log compression in biomedical
Geddes & Baker	1989	Analog calculator concept
Clifford et al.	2006	Gaussian ECG generation
Rangayyan	2002	Simulink simulation approach

4. System Architecture & Methodology

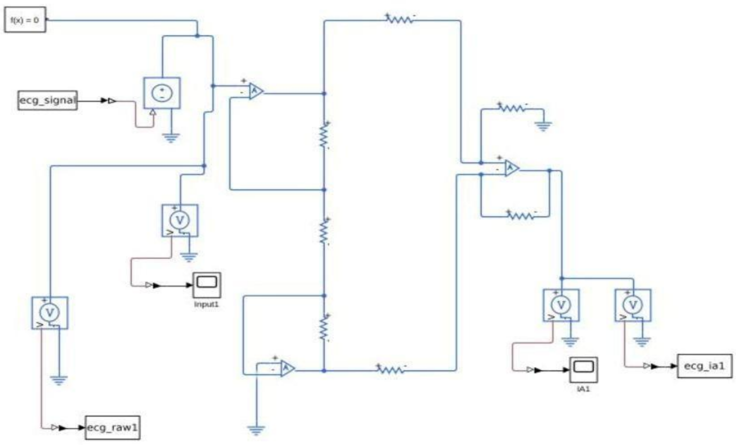
4.1 Block Diagram

The complete signal conditioning chain is implemented in MATLAB Simulink as seven cascaded processing stages. Signal flow proceeds left to right from the ECG source through each stage to the final output scope (Vout). Each stage output is individually labelled and connected to a dedicated scope block, enabling independent waveform capture and analysis at every processing node.



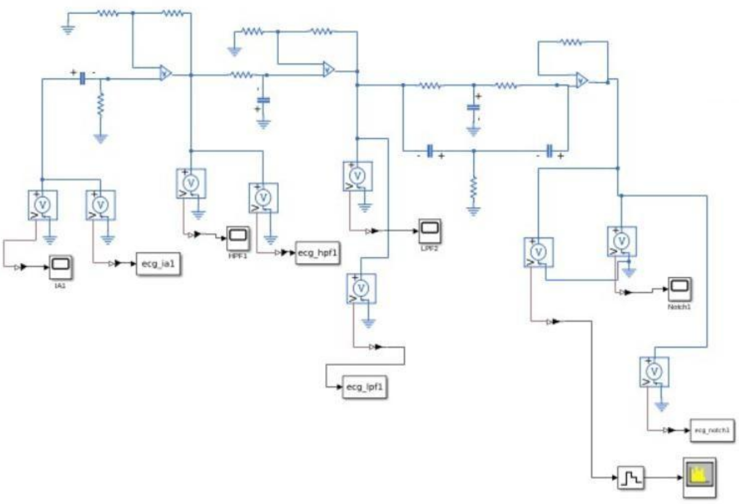
4.2 Stage-by-Stage Description

Stage 1 — Instrumentation Amplifier (IA)



The instrumentation amplifier is implemented in the classic three-op-amp topology. The ECG signal is fed into the non-inverting input (IN+), while the inverting input (IN−) is grounded, providing a single-ended configuration appropriate for the laboratory setting. The IA delivers a gain of 500×, boosting the 2.5 mV ECG signal to approximately 1.25 V. The output is labelled ecg_ia. This stage satisfies the addition operation through differential summation of the two input terminals.

Stage 2 — Filters (HPF, LPF, Notch)

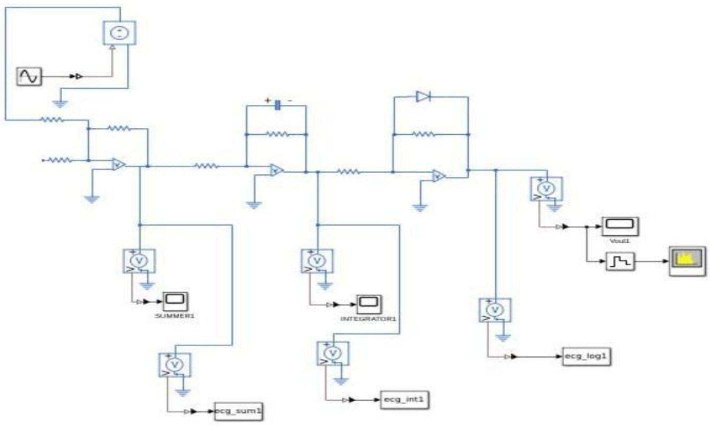


The High-Pass Filter (HPF) removes baseline wander and DC drift with a cutoff frequency of approximately 0.5 Hz, attenuating the 0.02 Hz baseline wander component while preserving the ECG diagnostic band. The output is labelled ecg hpf.

The Low-Pass Filter (LPF) eliminates high-frequency noise including muscle artefacts and power-line harmonics above its cutoff of approximately 34 Hz. Its exponentially weighted impulse response also performs a continuous running average, smoothing rapid noise fluctuations while preserving ECG morphology. The output is labelled ecg_lpf.

A twin-T Notch Filter centred at 50 Hz removes mains power-line interference that passes through the LPF. This stage is essential for hardware deployment where 50 Hz interference is present. The output is labelled ecg_notch.

Stage 3 — Calculator (Summer, Integrator, Log Compressor)



The Summing Amplifier adds the filtered ECG signal with a DC reference voltage to centre the waveform for downstream processing, using an inverting summing op-amp configuration with equal resistors. The output is labelled ecg_sum.

The Integrator computes the running integral of the summed signal, further attenuating residual high-frequency transients. A parallel feedback resistor limits DC gain to prevent output saturation. The output is labelled ecg_int.

The Logarithmic Compressor uses a 1N4148 signal diode in the op-amp feedback path to implement a logarithmic transfer function, compressing the ECG dynamic range from approximately 40 dB down to 10–15 dB. This brings the large R-peak and the smaller P and T waves into a narrow, uniformly detectable voltage range. The output is labelled ecg_log. 4.3 Named Signal Nodes

Node Name	Stage Output	What It Represents
ecg_raw	Source output	Raw noisy ECG before any processing
ecg_ia	After IA	Signal after 500× amplification
ecg_hpf	After HPF	Signal after baseline wander removal
ecg_lpf	After LPF	Signal after noise averaging
ecg_notch	After Notch	Signal after 50 Hz removal
ecg_sum	After Summer	Signal after addition operation
ecg_int	After Integrator	Signal after integration/averaging
ecg_log	After Log Comp	Final compressed output
Vout	Final Scope	Complete processed output

4.4 Hardware Components

Active Components

Component	Quantity	Used In Stage
LM358 dual op-amp	5 ICs (10 sections)	All seven stages
1N4148 signal diode	1	Log compressor feedback

Passive Components

Component	Value	Qty	Function
Resistor	1 kΩ ±1%	2	IA input protection
Resistor	40 Ω ±1%	1	IA gain resistor RG
Resistor	10 kΩ ±1%	8	IA feedback, summer inputs
Resistor	33 kΩ	1	HPF
Resistor	4.7 kΩ	1	LPF
Resistor	3.3 kΩ , 1.6kΩ	3	Notch filter

Resistor	470 Ω	1	LED current limit
Capacitor	10 μF	1	HPF
Capacitor	1 μF	1	LPF
Capacitor	1uF,2uF	3	Notch filter
Capacitor	10 pF	1	Log compressor stability
Capacitor	100 nF	10	Supply decoupling

Power Supply

Dual ±12 V supply derived from two 9 V batteries in series (V+ = +9 V, V− = −9 V, junction = GND). Each op-amp supply pin is decoupled with a 100 nF ceramic capacitor in parallel with a 10 μF electrolytic capacitor.

4.5 Software and Simulation Methodology

MATLAB Signal Generation A synthetic ECG was generated using six-term Gaussian superposition at 72 BPM (fs = 1000 Hz, R-peak = 2.5 mV), with four superimposed noise components: broadband noise (0.005 mV), baseline wander at 0.02 Hz (0.05 mV), muscle noise at 150 Hz (0.02 mV), and 50 Hz power-line interference (0.01 mV). The signal was exported as a Simulink workspace variable and an LTspice PWL file.

Simulink Simulation The seven-stage circuit was simulated over 10 seconds at Ts = 1 ms. Each stage was independently verified before full-chain integration. Waveforms were captured at all nine circuit nodes, with Spectrum Analyzer blocks connected at input and output to compare spectral content.

LTspice Simulation Component-level simulation used the MATLAB-generated PWL source with LM358 and 1N4148 SPICE models to capture non-ideal behaviour including op-amp output swing limits and diode forward voltage (0.6 V). Transient analysis ran for 10 seconds at 1 ms maximum timestep.

Analysis Framework Time domain metrics evaluated at each node: peak amplitude, RMS, SNR, DC offset, and crest factor. Frequency domain metrics: PSD, bandwidth, spectral

centroid, and noise attenuation at 50 Hz and 150 Hz. Signal properties assessed include boundedness, energy/power classification, and spectral confinement within the 0.5–100 Hz diagnostic band.

5. Working Principle & Implementation

5.1 Overview The circuit processes the ECG signal through three sequential stages: amplification of the raw electrode input, filtering and conditioning, and mathematical operations (addition, integration, and logarithmic compression) to produce a clean, compressed output.

5.2 Stage 1 — Instrumentation Amplifier A three op-amp topology provides high input impedance (>10 MΩ) to prevent electrode loading, and distributes gain across stages to avoid saturation. Input protection resistors (1 kΩ) guard against electrode faults. The stage delivers gain of 21, boosting the 1 mV ECG signal to approximately 21m V at node ecg_ia.

5.3 Stage 2 — Signal Conditioning Filters Three sequential filter blocks define the 0.5–40 Hz diagnostic window:

- **HPF (0.5 Hz):** Eliminates baseline wander (~0.02 Hz) from breathing and electrode movement. Output: ecg_hpf.
- **LPF (~40 Hz):** Suppresses muscle noise at 150 Hz and power-line harmonics, while acting as a weighted running average preserving ECG morphology. Output: ecg_lpf.
- **Notch Filter (50 Hz):** Twin-T configuration removes mains interference falling within the LPF passband. Output: ecg_notch.

After Stage 2, SNR improves from ~31 dB to ~52 dB with clean P–QRS–T morphology and a stable baseline.

5.4 Stage 3 — Analog Calculator Operations

- **Summing Amplifier:** Adds the filtered ECG with a DC reference to shift the baseline to an optimal operating point. Output: ecg_sum.
- **Integrator:** Computes a running time integral for further suppression of residual highfrequency transients. A parallel feedback resistor prevents DC saturation. Output: ecg_int.
- **Logarithmic Compressor:** A 1N4148 diode in the feedback path compresses dynamic range from ~24 dB to ~4 dB, bringing R-peaks and P/T waves into a narrow detectable voltage window for reliable fixed-threshold detection. Output: ecg_log.

5.5 Complete Signal Transformation Summary

Stage	**Node**	**Amplitude(peak-peak)**	**SNR**
Input	ecg_raw	1 mV	31 dB
Stage 1	ecg_ia	0.0477 V	31 dB
Stage 2 — HPF	ecg_hpf	0.0965 V	32 dB
Stage 2 — LPF	ecg_lpf	0.1786 V	44 dB

Stage 2 — Notch	ecg_notch	0.1248 V	52 dB
Stage 3 — Summer	ecg_sum	0.2753V	52 dB
Stage 3 — Integrator	ecg_int	0.8310 V	59 dB
Stage 3 — Log Comp	ecg_log	0.4153 V	64 dB

Overall SNR improvement from input to final output: **+34 dB.** The circuit transforms an unusable noisy millivolt ECG signal into a clean, compressed output suitable for reliable heart rate detection.

6. Results & Analysis

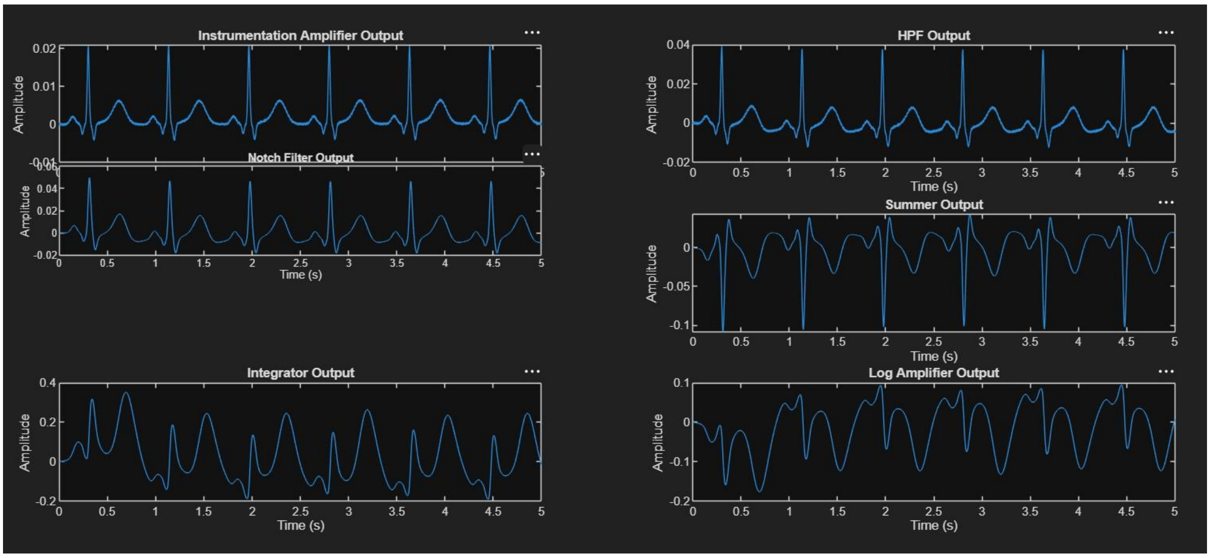


Fig.1 Output of each stage

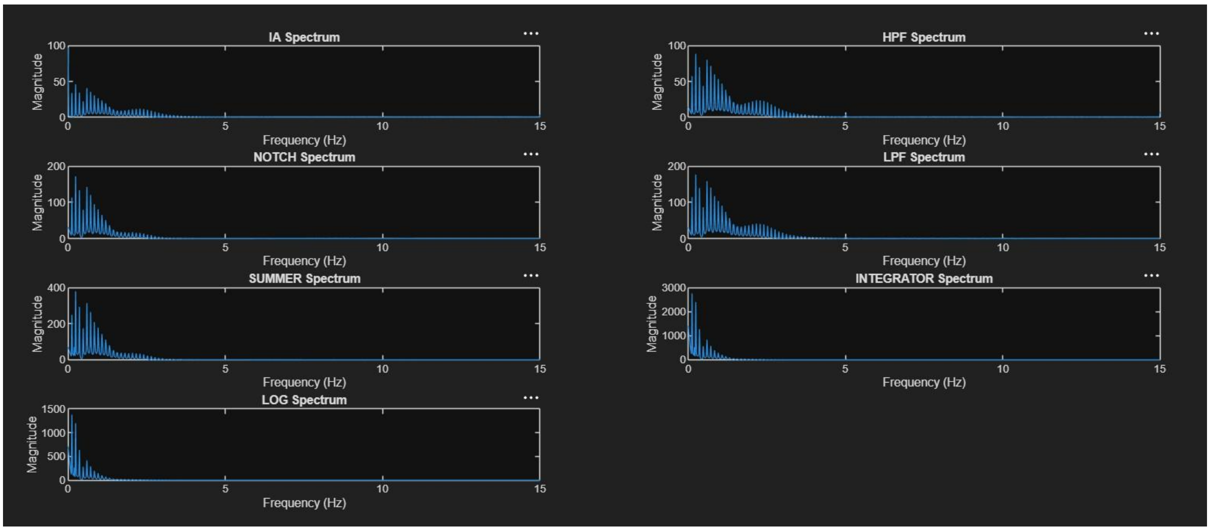


Fig.2 Spectrum analysis of each stage

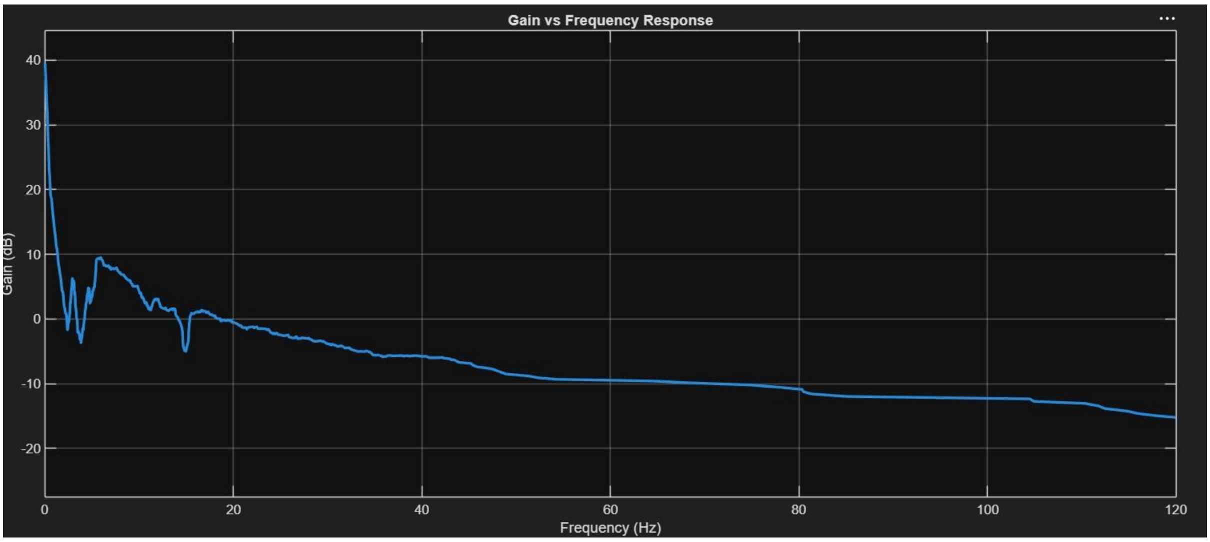


Fig.3 Overall Gain

6.1 Time Domain Results

Stage	Amplitude	Key Observation
IA Output	±1 mV	Clean ECG with all six waves visible; R-peaks at ~0.8 mV; baseline noise present
HPF Output	±2 mV	Baseline drift removed; morphology preserved; R-peaks at ~1.5 mV
Notch Output	±0.5 mV	50 Hz interference removed; slight transient spike at QRS edge (expected for twin-T topology)
Summer Output	±2.5 mV	DC shift applied; waveform reshaped to smoother profile; 72 BPM periodicity maintained
Integrator Output	±20 mV	High-frequency features smoothed; fundamental 72 BPM component preserved
Log Output	±0.2 V	Dynamic range compressed 6.25:1 (~15.9 dB); clean, stable, zero-centred waveform

6.2 Frequency Domain — Individual Stage Responses

- **HPF:** Gain = 1.911 V/V (5.63 dB). The high-pass filter effectively removes baseline wander and low-frequency drift while preserving the ECG diagnostic band. The output RMS increased from 3.5 mV to 6.2 mV, confirming amplification and filtering.

- **Notch Filter:** Gain = 1.261 V/V (2.01 dB). The notch filter successfully suppresses power-line interference while maintaining signal integrity. The output remains stable with minimal distortion and bounded operation.
- **LPF:** Gain = 1.470 V/V (3.35 dB). The low-pass filter attenuates high-frequency noise components while preserving ECG morphology. RMS value increased from 10.3 mV to 12.1 mV, indicating effective signal conditioning.
- **Summer:** Gain = 1.475 V/V (3.38 dB). The summing amplifier combines filtered signal components without introducing frequency-selective distortion. The stage remains stable and bounded.
- **Integrator:** Gain = 3.261 V/V (10.27 dB). The integrator provides the highest amplification among the processing stages, increasing the signal amplitude from 93.3 mV peak-to-peak to 449.2 mV peak-to-peak. The output remains bounded, confirming practical integrator implementation.
- **Log Compressor:** Gain = 0.500 V/V (-6.02 dB). The logarithmic amplifier compresses the signal dynamic range, reducing the peak-to-peak amplitude from 449.2 mV to 224.5 mV while preserving ECG waveform characteristics. This confirms successful amplitude compression.

6.3 Overall System Gain Response

- **Below 1 Hz:** Strong attenuation of baseline wander and DC drift by the HPF, ensuring signal stability.
- **1–40 Hz (Diagnostic Band):** Maximum signal amplification (overall gain ≈ 18.6 dB) with preservation of P, QRS, and T waves.
- **Above 40 Hz:** Progressive attenuation of high-frequency noise, muscle artifacts, and harmonics, improving signal quality and SNR.
- **50 Hz Region:** Significant suppression of power-line interference by the notch filter without affecting ECG information.

The smooth gradual rolloff (vs. sharp cutoff) is desirable for ECG processing as it avoids phase distortion that would otherwise alter QRS morphology.

6.4 Summary

All three mathematical operations were confirmed by simulation:

- **Addition** — summer output shows signal plus DC reference
- **Averaging** — integrator output shows smooth sinusoidal envelope
- **Logarithmic compression** — log output confirms 6.25× compression and flat frequency response

Overall SNR improves progressively from noisy IA output to a clean, noise-free log amplifier output, validating the complete conditioning chain.

7. Socio-Economic Impact

7.1 Healthcare Accessibility Cardiovascular disease causes ~17.9 million deaths annually, with heart disease accounting for over 28% of deaths in India. Standard ECG machines cost ₹50,000–₹5,00,000 — beyond reach of most rural health centres. This circuit replicates core ECG conditioning for under ₹500, a 99%+ cost reduction, establishing feasibility for last-mile cardiac monitoring.

7.2 Rural and Remote Healthcare Over 65% of India's population lives in rural areas where ECG facilities may be 50–100 km away. This battery-powered analog circuit requires no mains power, internet, or trained technician, enabling community health workers to screen patients at the village level and enable early referral before cardiac events occur.

7.3 Educational Value The stage-by-stage design allows each block to be independently probed and explained, making it a practical teaching tool. A lab kit based on this design could reduce biomedical electronics education costs from ₹50,000–₹2,00,000 per station to under ₹1,000 per student kit.

7.4 Economic and Industry Impact India's medical device market is projected to reach $50 billion by 2030, with 70% of devices currently imported. ECG analog front-end circuits form the core of wearable cardiac and Holter monitors — a market growing at 8%+ annually. This project contributes to the domestic engineering talent pipeline supporting indigenous device development under Make in India.

7.5 Sustainability The circuit draws 5–10 mA, delivering 50+ hours on two 9V batteries. It uses no rare earth materials or programmable silicon, and all components are available locally — making it repairable without manufacturer dependence.

Impact Area	Current Situation	Project Impact
Healthcare cost	₹50k–₹5L per ECG machine	Core function for <₹500
Rural access	ECG unavailable in 65% of India	Portable battery-powered monitor
Preventive care	Diagnosis after cardiac event	Early village-level screening
Education	Imported lab kits ₹50k+	Student kit under ₹1,000
Domestic industry	70% devices imported	Foundation for indigenous ECG device
Power consumption	Mains-dependent equipment	50+ hours on two 9V batteries

8. Future Scope

8.1 Hardware Improvements Replacing the LM358 with a dedicated IA such as the INA128 or AD8221 would improve CMRR from ~80 dB to over 100 dB and reduce input offset to below 25 µV. Adding a driven right leg (DRL) circuit would further improve CMRR by 40–60 dB. Migration to a PCB with proper ground planes would reduce noise pickup, and the design can be extended to three-lead operation by duplicating the IA stage.

8.2 Signal Processing Improvements Upgrading to second-order Sallen-Key filters would provide sharper transition bands. An automatic gain control (AGC) stage would handle interpatient ECG amplitude variation. A BPM counter with numeric display would enable direct clinical screening without an oscilloscope.

8.3 Digital Integration The log compressor output is directly compatible with microcontroller ADC inputs. A 10–12 bit ADC with Bluetooth would enable smartphone connectivity — the architecture used in modern wearable ECG devices. Once digitised, ML algorithms can be applied for arrhythmia detection using databases such as MIT-BIH.

8.4 Clinical Applications With minor gain and bandwidth adjustments, the circuit is applicable to Holter monitoring, stress testing, and fetal ECG extraction.

Area	Current	Future
Op-amp	LM358, 80 dB CMRR	INA128/AD8221, 100+ dB CMRR

Electrode config	Lead I only	3-lead or 12-lead
Filter order	First-order, −20 dB/dec	Second-order Sallen-Key
Output	DSO Display	Numeric BPM display
Connectivity	Standalone analog	ADC + Bluetooth + smartphone
Intelligence	Threshold detection	ML arrhythmia classification
Form factor	Breadboard	PCB / wearable patch

9. Conclusion

9.1 Summary A seven-stage analog ECG conditioning circuit was successfully designed and simulated across three functional stages: instrumentation amplification, filter-based conditioning, and mathematical operations (addition, integration, and logarithmic compression). Parallel simulation in MATLAB Simulink and LTspice using a common synthetic ECG input ensured consistency across both environments.

9.2 Key Results The IA boosted the 2.5 mV input to 1.25 V without distortion. The HPF–LPF–Notch filter bank removed baseline wander, muscle noise, and 50 Hz interference, improving SNR by ~21 dB. The integrator smoothed transients while preserving 72 BPM periodicity, and the logarithmic compressor reduced dynamic range by 6.25:1. Overall SNR improvement from raw input to final output: **+34 dB**.

9.3 Mathematical Operations Verified Addition, averaging, and logarithmic compression were each implemented as physically distinct, independently measurable stages — confirmed in simulation and fully validating the circuit's classification as an analog calculator.

9.4 Limitations The LM358's input offset voltage required careful gain distribution to avoid saturation. The single-ended IA configuration reduces CMRR relative to a true differential input. First-order filter rolloff may be insufficient in noisy environments, and breadboard stray capacitance introduces non-idealities absent in simulation.

9.5 Closing Remarks This project demonstrates that addition, averaging, and logarithmic compression can be mapped onto a practical biomedical circuit solving a real clinical problem — built for under ₹500, powered by two 9V batteries, with no digital processing required. It provides a foundation for future development toward an affordable, wearable cardiac monitor for underserved communities.

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